

Data Mining Project

Name: Waleed Hamoud Al-Harbi

ID: 441008303

Abstract

In this report, I analyzed six machine learning tasks: four classification tasks and two clustering tasks. For the classification part, I compared Decision Tree, KNN, and Naive Bayes on four different datasets. For the clustering part, I used K-Means and relied on the elbow method to choose a suitable number of clusters. I also included a short description for each dataset and summarized the main results in tables. In general, the classification results were stronger on the simpler datasets, while the first clustering task showed a clearer clustering pattern than the second one.

1. Introduction

This project includes six separate tasks. The first four are supervised classification tasks, and the last are unsupervised grouping tasks. My main goal in this report is to explain the data sets, compare the model results, and present the final results in a clear and structured way.

While working through the analysis, I pre-processed the data, which is important because it affects how the data is read and how the final results are explained.

2. Methodology

To prepare this report, I went through all six tasks carefully and checked that the structure of the dataset matched the variables used in the code. The printed results were also compared with the evaluation steps to ensure that the final figures were consistent.

For the four classification tasks, the same general workflow was followed: selection of input features and the target variable, splitting the data into training and test groups using an 80/20 ratio with stratification, scaling the data for BE and Naive Bayes when needed, training the three models, and then evaluating them using confusion matrices with accuracy, precision, recall and Formula 1 score.

For the two clustering tasks, one column was removed before clustering was applied (charges in clustering 1 and emissions in clustering 2). After that, the remaining variables were scaled, k-means were tested with K-values from 2 to 5, and the elbow method was used by comparing the inertia values.

3. Dataset Descriptions

One of the project requirements was to give a short description for every dataset and to show the main classification and clustering results in tables. Table 1 gives a quick summary of the six datasets.

Dataset	Task	Rows	Columns	Main Features	Target Column	Missing	Duplicates
Classification 1	Classification	484	3	StudyHours, Attendance	Pass	0	0
Classification 2	Classification	380	4	Gender, Age, EstimatedSalary	Purchased	0	0
Classification 3	Classification	2000	21	20 technical phone features	price_range	0	0
Classification 4	Classification	764	9	8 medical features	Outcome	0	0
Clustering 1	Clustering	1337	10	age, sex, bmi, children, smoker, region *	charges (excluded)	0	0
Clustering 2	Clustering	18967	114	vehicle, fuel, transmission and encoded fields	emissions (excluded)	0	0

Classification 1

This dataset is about student performance. It has 484 rows and 3 columns, and it uses StudyHours and Attendance to predict Pass. The two classes are close in size, with 230 records in class 0 and 254 records in class 1, so it is a fairly balanced dataset for classification.

Classification 2

This dataset has 380 rows and 4 columns. From the variables, it looks like a customer purchase prediction dataset that uses Gender, Age, and EstimatedSalary to predict Purchased. The target is slightly imbalanced toward class 0, but the dataset is still clean and suitable for a basic model comparison.

Classification 3

This dataset has 2,000 rows and 21 columns. It seems to be a mobile phone price dataset with 20 input features and a four-class target called price_range. Each class contains 500 records, so the dataset is perfectly balanced. The values also look as if they were already scaled before the analysis stage.

Classification 4

This is a medical dataset with 764 rows and 9 columns. It uses eight health-related variables to predict diabetes outcome. The data is clean, but the target is not balanced because class 0 appears more often than class 1.

Clustering 1

This dataset has 1,337 rows and 10 columns. From the variables, it looks like an insurance or customer-profile dataset. The analysis removes charges before clustering and groups the records using features such as age, sex, BMI, children, smoker, and region indicators. Since the variables are on different scales, normalization is important here.

Clustering 2

This dataset contains 18967 rows and 114 columns. From feature names to vehicle data set and fuel consumption after heavy coding. The analysis eliminates emissions before collection.

4. Classification Results

The next tables summarize the classification results. For each dataset, I compared Decision Tree, KNN, and Naive Bayes, then highlighted the model that achieved the best performance.

Classification 1

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.8144	0.8146	0.8144	0.8142
KNN	0.8041	0.8132	0.8041	0.8014
Decision Tree	0.7320	0.7339	0.7320	0.7301

On this dataset, Naive Bayes gave the highest accuracy, 0.8144. The difference between Naive Bayes and KNN was small, but Naive Bayes still came first. Since the dataset only has two simple input variables, a simple probabilistic model was enough to perform well.

Confusion matrix for the best model (Naive Bayes):

Actual	Pred 0	Pred 1
Class 0	36	10
Class 1	8	43

Classification 2

Model	Accuracy	Precision	Recall	F1-Score
KNN	0.9079	0.9125	0.9079	0.9088
Naive Bayes	0.8816	0.8827	0.8816	0.8820
Decision Tree	0.8158	0.8194	0.8158	0.8170

Here, KNN was the strongest model with an accuracy of 0.9079. This suggests that nearby records in the feature space were very similar in terms of the purchase decision, so the distance-based method worked well.

Confusion matrix for the best model (KNN):

Actual	Pred 0	Pred 1
Class 0	43	5
Class 1	2	26

Classification 3

Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.8425	0.8461	0.8425	0.8429
Naive Bayes	0.8100	0.8113	0.8100	0.8105
KNN	0.5200	0.5398	0.5200	0.5242

For this dataset, Decision Tree achieved the best result with an accuracy of 0.8425. KNN performed much worse than the other two models. I think this happened because the dataset has many features and four classes, which makes distance-based classification harder.

Confusion matrix for the best model (Decision Tree):

Actual	Pred 0	Pred 1	Pred 2	Pred 3
Class 0	92	8	0	0
Class 1	14	77	9	0
Class 2	0	11	84	5
Class 3	0	0	16	84

Classification 4

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.6928	0.6915	0.6928	0.6921
KNN	0.6797	0.6708	0.6797	0.6738
Decision Tree	0.6601	0.6544	0.6601	0.6568

Naive Bayes gave the highest accuracy on this dataset, 0.6928, but the three models were fairly close to each other. Compared with the earlier datasets, this task was more difficult, which suggests more overlap between the two classes.

Confusion matrix for the best model (Naive Bayes):

Actual	Pred 0	Pred 1
Class 0	77	23
Class 1	24	29

Best Model Summary

Dataset	Best Model	Accuracy	F1-Score
Classification 1	Naive Bayes	0.8144	0.8142
Classification 2	KNN	0.9079	0.9088
Classification 3	Decision Tree	0.8425	0.8429
Classification 4	Naive Bayes	0.6928	0.6921

5. Clustering Results

The clustering results are summarized using inertia values from the elbow method. A lower inertia means the points are closer to their cluster centers, but the key idea is to look for the point where the improvement starts to slow down.

Clustering 1

k	Inertia
2.0	10110.3232
3.0	8332.3289
4.0	6565.6468
5.0	6142.4396

The first clustering task shows a clearer elbow than the second one. The inertia drops a lot from $k=2$ to $k=3$ and again from $k=3$ to $k=4$, but the improvement becomes much smaller at $k=5$. Because of that, $k=4$ looks like the most reasonable choice from the tested values.

Clustering 2

k	Inertia
2.0	2345389.1200
3.0	2299465.6700
4.0	2272540.5800
5.0	2244677.1800

The second clustering task does not show a sharp elbow. The biggest improvement appears when moving from $k=2$ to $k=3$, while the later changes are small and close to each other. This means the natural cluster structure is weaker here, so $k=3$ can be used as a practical starting point, but the result should be read with caution.

6. Discussion

When the datasets were simple, clean and low-dimensional, the models performed very well. This can be clearly seen in classification 1 and Classification 2, where Naive Bayes and Ken achieved solid results. Classification 3 was different because it had many features and four categories. In this case, the decision tree coped with the data structure much better and achieved the highest accuracy. KN was much weaker, matching the common problem that distance-based models become less effective in more complex feature spaces. Classification 4 was the most difficult of the four classification tasks. The best accuracy was only about 69%, which indicates that the two categories overlap more and the available features do not clearly separate them. In the grouping part, the first data set gave a more readable elbow pattern, while the second data set showed a weaker separation.

7. Conclusion

After reviewing all six tasks, the rating results can be summarized as follows: Naive Bayes was the best in rating 1 and rating 4, Ken was the best in rating 2, and the decision tree was the best in rating 3. For assembly, assembly 1 showed a reasonable attachment around which $K = 4$, while assembly 2 showed a weaker structure and was difficult to clearly separate.

In general, the project is organized in a clear manner, and the frequent evaluation steps have made it easy to compare the results across datasets.